

Getting started with Dart



Dart

Speed is of the essence here

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An up and coming language that has the potential to blow up in popularity even more is Dart, a programming language by Google. It's Ahead Of Time (AOT) that compiles to native code which makes it extremely fast and practically everything written in it can be easily customised. It's most popularly used in Flutter which a UI framework, also by Google. The good thing about Dart is that it can also be Just In Time (JIT), so you can rapidly iterate without having to spend ridiculous amounts of time sitting around waiting for the code to finish compiling. Flutter's sub-second stateful hot reload feature is made possible by this aspect.

The whole aspect of Dart being compiled into native code means that it's much quicker to execute. This also means that whatever UI elements are created with flutter will be smoother during execution. It can perform object allocation and garbage collection without using locks. It doesn't perform

preemptive scheduling or use shared memory. Apps built in Dart thus appear to execute faster.

Then there's the fact that because Dart methods and syntax is familiar to both, static and dynamic languages, it's easier to pick up for everyone. And you don't need to have a declarative language such as XML because the declarative and programmatic layout within Dart is easy to work with. By utilising fewer third-party tools, developers might find the workflow to be less cumbersome. Also, we should mention that Dart is open-source.

INSTALLING DART

If you've used any package managers, then installing Dart is super easy. You can use Chocolatey to install the Dart SDK. Chocolatey is very similar to npm and installing it seems more of a bother than installing the Dart SDK. To install chocolatey, type the following in your command prompt window:

```
@%SystemRoot%\System32\WindowsPowerShell\v1.0\powershell.exe" -NoProfile -InputFormat None -ExecutionPolicy Bypass -Command "iex ((New-Object System.Net.WebClient).DownloadString('https://chocolatey.org/install.ps1'))" && SET "PATH=%PATH%;% ALLUSER-SPROFILE%\chocolatey\bin"
```

Once you have Chocolatey installed, you can install the Dart SDK. Simply type the following in the CLI:

```
Choco install dart-sdk
```

It would be ideal to also install stagehand which has templates that allow you to quickly create Dart apps. Install stagehand by typing the following in the command line:

```
pub global activate stagehand
```

Stagehand has several goto templates that might come in handy. The following are often used:

- console-full** - A command-line application sample.
- package-simple** - A starting point for Dart libraries or applications.
- server-shelf** - A web server built using the shelf package.
- web-angular** - A web app with material design components.
- web-simple** - A web app that uses only core Dart libraries.
- web-stagexl** - A starting point for 2D animation and games.

Once you have Stagehand installed, you can move onto creating your app using one of the templates that are now available to you.

```
mkdir cli
cd cli
stagehand console-full
```

You should now have a file within your binary folder named main.dart. This is where you'd start off with the app. An additional file in the lib folder named cli.dart will also be generated. This is imported by your main.dart file and contains most of the functionalities of your app. Lastly, a metadata file